

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #6 (Lesson 21) - Array Factor Lab
Lab Sheet
Fall 2026

Name: _____ Section: _____ Date: _____

Learning Objective 3.2: Derive and apply the array factor for a uniform linear array, and relate its beamwidth and sidelobe structure to what a beam sweep measures.

This sheet is the turn-in document for the Lesson 21 lab. Fill the calculated columns before you sweep and the measured columns at the bench, so the prediction is on paper before the trace is on the screen. The lab is a team assignment, so one sheet per team.

1. **The measurement table.** One row per aperture, at 10.3 GHz with $d = 14$ mm. Report each trace’s peak level, and the step from the eight-element peak beside it, and report each sidelobe level relative to that trace’s own peak.

Active elements	Peak (dBFS)	HPBW meas	HPBW calc	FNBW meas	FNBW calc	First SLL
8 (Rx1-Rx8)						
4 (Rx3-Rx6)						
2 (Rx4, Rx5)						

2. **Written answer 1.** Explain why the first-null beamwidth of the two-element array is quoted as 180° rather than measured. Name the condition on $\lambda/2d$ that decides whether a null exists in visible space, and evaluate it for $d = 14$ mm at 10.3 GHz.

3. **Written answer 2.** The gain of each patch element is the same whether the other seven elements are on or off, and the amplifier behind each element is unchanged. Explain why the peak of the sweep nonetheless drops by about 6 dB every time you halve the number of active elements.

Documentation: _____